

Internet Computing

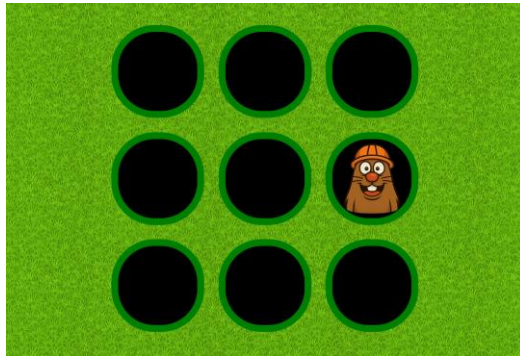
Group Project
Multi-player Game

Gibson Lam

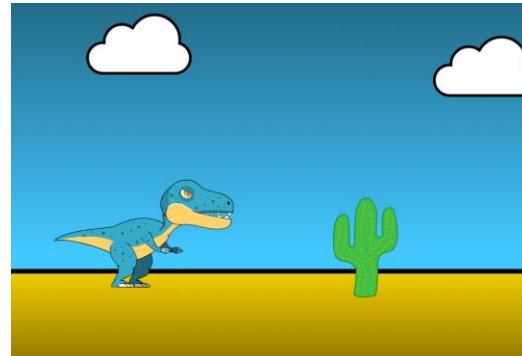
First Three Labs

- You created single-player games in the first labs using client-side techniques

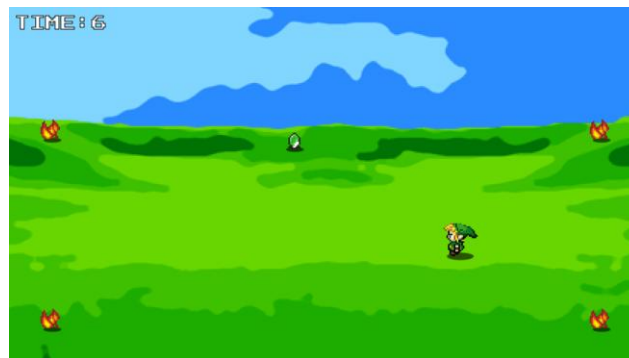
Lab 1



Lab 2



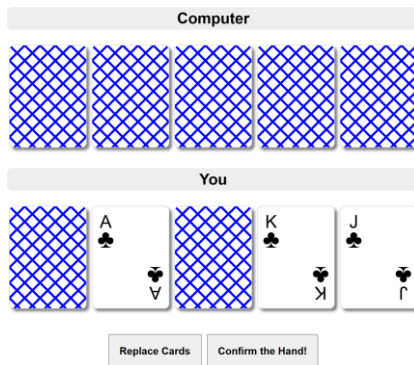
Lab 3



Next Three Labs

- In the next labs, you will build games (and a game room) using server-side techniques

Lab 4



The Game Room

Sign In Register

Lab 5

Rock, Paper, Scissors



Lab 6
(to be released)

The Group Project

- In the group project, you will develop a multi-player game with a web server
- Loosely speaking, your tasks involve combining the functionalities of the labs into one single new application



A multi-player
card game

Forming a Group

- You will form a group of **2 to 3 people**
- Get together with your members and add yourself to make a group here:

<https://canvas.ust.hk/courses/67702/groups>

Making Your Group

- If you are looking for group members, feel free to do it here:

[https://canvas.ust.hk/courses/67702/
discussion_topics/601234](https://canvas.ust.hk/courses/67702/discussion_topics/601234)

- If you are not in a group after the deadline shown below, you will be put in a group
- The deadline for making a group is

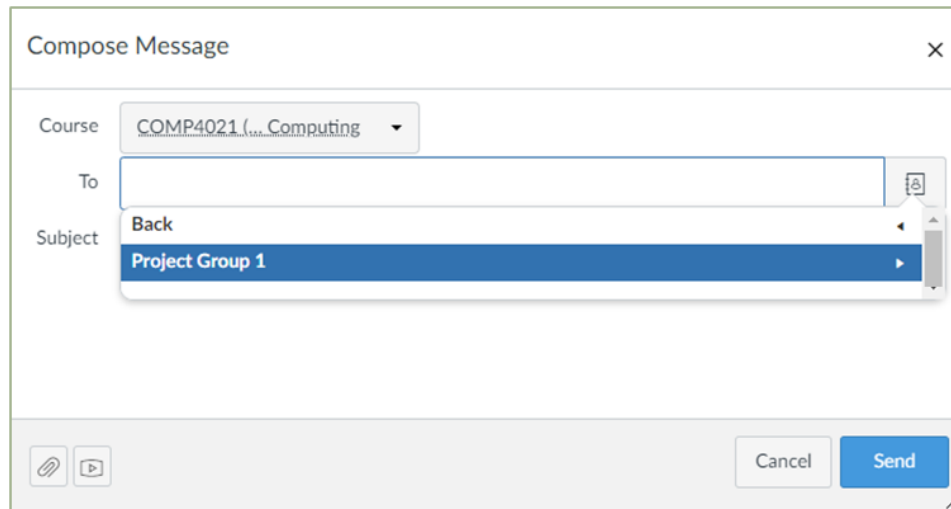
11:59pm, Monday, 13 Apr 2026

Group Home Page

- Each group has a 'group homepage' in the canvas system
- You can access your group homepage via People > Groups
- In the project area you can communicate with your group, if you want to do that
- The instructor can also access your group area, but students outside your group cannot do that

Messaging your Group

- You can use the canvas mail system to send messages to your group

A screenshot of the 'Compose Message' dialog box in Canvas LMS. The dialog has a title bar with a close button (X). It contains a 'Course' dropdown menu set to 'COMP4021 (... Computing)'. Below this is a 'To' field with a contact icon on the right. The 'Subject' field is open, showing a dropdown menu with 'Back' and 'Project Group 1' (which is highlighted in blue). At the bottom left are icons for attaching files and videos. At the bottom right are 'Cancel' and 'Send' buttons.

- However, using Whatsapp or a similar app would probably be more efficient

Topic of the Game

- You can design your own game for the group project, which does not have to be something that you have done in the course
- For example, you can make:
 - A shooting game
 - A puzzle game
 - A platform game
 - A board game

Shooting Games



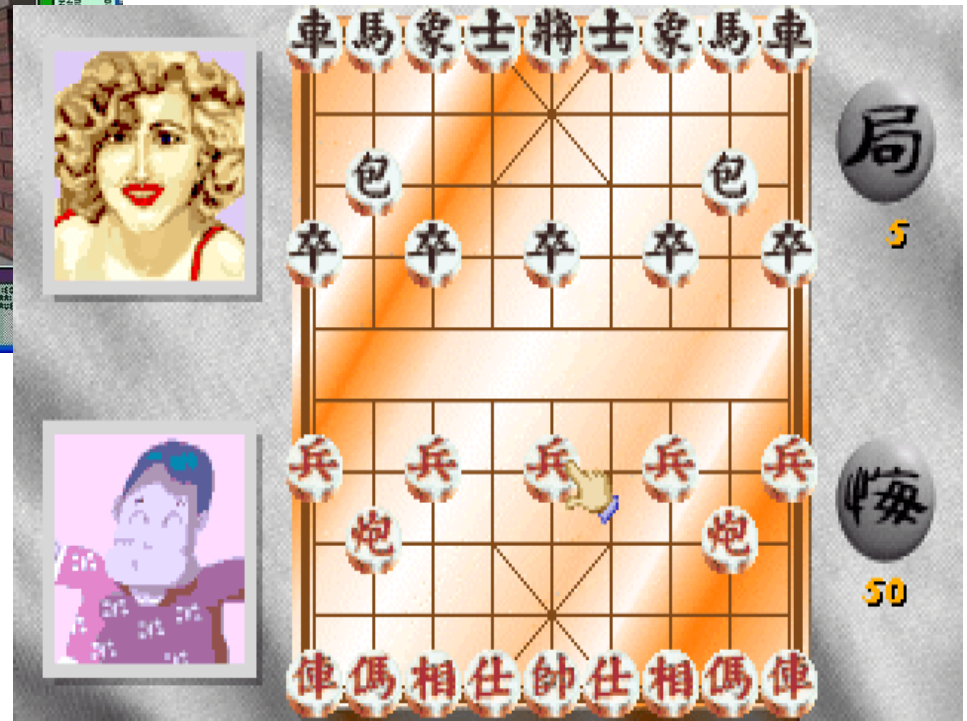
Platform Games



Puzzle Games



Board Games

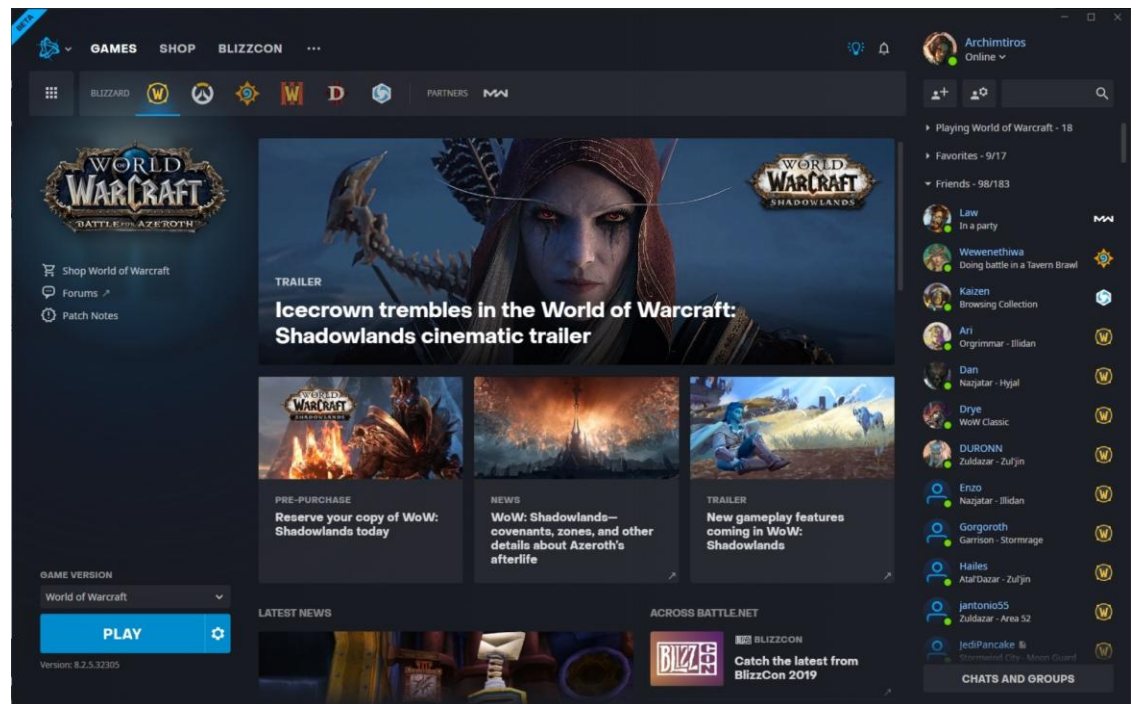


Components of the Game

- Your multiplayer game should have these components:
 - The game home page
(e.g. based on the work in lab 5 and 6)
 - The game play page
(e.g. based on the work in some of the other labs)
 - The game over page

Game Home Page

- This is the front page of the game web application, which should include the description and instructions on how to play the game
- Players can **register**, **sign-in** and **pair up** for the game here



Game Play Page

- This is where the actual game takes place
- The game must be a **two-player game**
- Interactions between the two players should be shown **in almost real-time** using WebSocket API



Things in the Game

- There must be at least **three different types** of elements in the game
- One of them must be **the players**
- The others can be enemies, obstacles, cards, and so on



Game Controls

- You can choose the game control method
- For example, you can use keyboard keys:

```
$(document).on("keydown", ...
```

- And/or you can use mouse events:

```
$("#shoot-button").on("click", ...
```

Game Duration

- Normal game play should last for **at most 3 to 4 minutes**
- This helps to make sure the TAs can finish your game quickly!



Cheat Mode

- The game must include a **cheat code** or **cheat mode** so that a player can win easily
- The cheat mode can be activated by a secret key or button
- Please avoid using crazy combinations of keys!



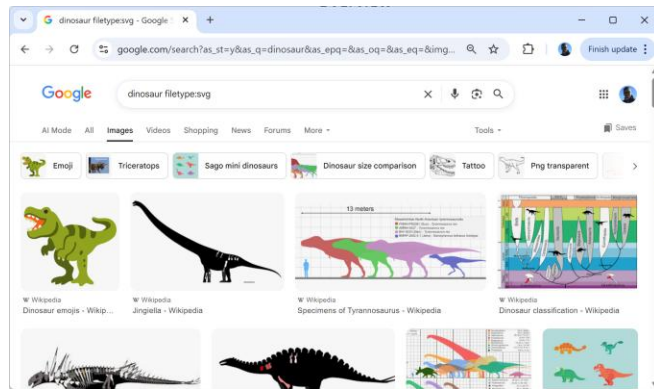
Game Over Page

- The game over page must contain statistics about the players
- An overall ranking of the players must also be shown



Graphics and Sounds

- The game play is created using either SVG or canvas – that means a **2D game**



- Your game can have different sound effects and background music using the HTML `<audio>` tag

Other Requirements

- You are expected to *primarily* use the codebase from your labs
- You **cannot** use any client-side game libraries or external frameworks
- You must use **Node.js** as the game web server (a single server)
- You may use images and sounds found on the web, or generated using AI

Running Your Game

- You may install additional packages, but we will run your game in the same way as the labs:
 1. Run `npm install` to install packages
 2. Start the server JavaScript file
 3. Browse to `http://localhost:8000`
- We will not use any scripts to automate the process

Submit Your Project Details

- Use the 'Announcement' feature in your canvas group area to announce:
 1. Title of your game
 2. Game home page
 - What will be in it
 3. Game play page
 - Describe the game, how to play it, how to win/lose
 4. Game over page
 - What will be in it
 5. Cheating
 - How you will support enabling and disabling it

Submit Your Project Details

- In your canvas group area, you can use the 'Announcement' feature to announce things
- Whatever you announce can be accessed by group members and the instructor
- It's a good idea to check the marking scheme before you do your announcement
- You need to announce the required information (previous slide) before:

11:59pm, Friday, 17 Apr 2026

Marking Overview 1/2

• Game home page		40
• Game description and instructions		5
• Player registration		10
• Player sign-in		10
• Player pair up		15
• Game play page		85
• Things in the game		30
• Players' interaction		20
• Game controls		10
• Game duration		10
• Cheat mode		15

- *All numbers shown are maximum amounts*
- *For each item, your project may receive anywhere from 0 up to the maximum score*

Marking Overview 2/2

- Game over page **30**
 - Player statistics 10
 - Player ranking 10
 - Restart the game/ back to front page 10
- Graphic and sounds **10**
- Running the project **15**
- Multi-user support **30**
- Game quality **50**
- Project details announcement **20**
- Project video **20**

Total maximum score = 300

Game Home Page 1/2

- **Game description and instructions** **5**
 - The game description and instructions are clearly shown on the home page
 - They should be properly written with enough details
- **Player registration** **10**
 - New players can register for a new account in the home page
 - The registration form should have proper data validation

Game Home Page 2/2

- **Player sign-in** **10**
 - Players can sign in the home page with an account
 - The sign-in form should have proper verification and error handling
 - Login session is maintained in the browser
- **Player pair up** **15**
 - Players can join a game with another player
 - Proper pairing mechanism should be shown in the interface

Game Play Page 1/2

- Things in the game **30**
 - There are at least 3 types of 'things' in the game
 - For example, monsters in a shooting game or cards in a card game are considered one type of thing
 - Any collection of the same thing should be animated / and properly interacted to be counted as one type of thing; static objects / text do not count
- Players' interaction **20**
 - Some kind of players' interaction is shown in almost real-time, instead of purely turn-based
 - WebSocket is used for the real-time communication

Game Play Page 2/2

- Game controls **10**
 - Some things can be effectively controlled by the mouse and/or keyboard
- Game duration **10**
 - A normal game should not last for more than 3 - 4 minutes
- Cheat mode **15**
 - A reasonably simple key/button can be used to activate the cheat mode
 - The game can be completed **much quicker** once the cheat mode is on (for example, unlimited bullets/ greater power/ ability to swap cards, etc)

Game Over Page

- Player statistics **10**
 - Relevant and useful statistics of the game play is shown for each player
- Player ranking **10**
 - Overall ranking of the players of some useful measures is shown
- Restart the game/ back to home page **10**
 - The game can be quickly restarted in this page
 - Or, the player can go back to the home page

Others 1/3

- Graphics and sound **10**
 - Game play uses SVG or canvas
 - At least two different sounds (i.e. 2 sound files) are used during game play
- Running the project **15**
 - A single 'npm install' can install all package dependencies
 - A single line of instruction starts the server
 - The game is started using `http://localhost:8000`
 - Please put a README html file in the submission if you need to make things extra clear

Others 2/3

- **Multi-user support** **30**
 - This refers to how well your project handles users
 - If the project can only handle one person at a time (turn-based 'monopoly' style), it may get 10 marks
 - But if the project can handle multiple people at the same time, in real time, it may get 30 marks
- **Game quality** **50**
 - This score is awarded based on overall creativity, completeness and playability
 - It is a general score given relative to the quality of work across the entire class

Others 3/3

- **Project details announcement** **20**
 - You announce the project details with sufficient details in your canvas group page before the deadline (see slide 25-26)
- **Project video** **20**
 - A video that clearly demonstrates all functionalities and features of your game
 - It should last for 5 minutes or less, with the majority of the video focused on gameplay

Submission

- Project files
 - Include:
 - HTML files, CSS files, JavaScript files, etc - all files needed
 - A package.json file
 - An optional README file e.g. readme.html
 - Put everything in a zip file called **<your group number>_project_files.zip**
 - Submit it through canvas
- Project video
 - A single MP4 video file called **<your group number>_project_video.mp4**
 - Submit it through canvas
- Deadline

11:59pm, Saturday, 9 May 2026

Summary of Deadlines

- Forming your group

11:59pm, Mon 13 Apr

- Project details announcement

11:59pm, Fri 17 Apr

- Project files and videos

11:59pm, Sat 9 May